

THE STRUCTURE OF MATHEMATICAL REALITY

Phase III — Chapter 7

Groups: The Language of Symmetry

Chapter 7 — Phase III: The Grammar of Structure

Self-Study Curriculum

Where This Sits

- Chapter 5 gave symmetry as an intuition: transformations that leave something unchanged.
- This chapter turns that word into an object you can compute with.
- A group is the exact formalization of symmetry.
- Payoff: crystals, equations, card shuffles, and conservation laws become one structure.

The Four Axioms

A group is a set G with an operation $\cdot$ such that:

1. **Closure:** $a \cdot b$ is in G .
2. **Associativity:** $(a \cdot b) \cdot c = a \cdot (b \cdot c)$.
3. **Identity:** an e with $e \cdot a = a \cdot e = a$.
4. **Inverses:** each a has a^{-1} with $a \cdot a^{-1} = e$. > Read them as one thing: reversible operations that can be composed.

What a Group Captures

- The transformations that leave a property unchanged always satisfy the four axioms.
- The axioms are not imposed on symmetry; they are what symmetry automatically is.
- molecule $\rightarrow$ spectroscopy; equation $\rightarrow$ solutions; spacetime $\rightarrow$ the Lorentz group.
- permutations $\rightarrow$ combinatorics and cryptography; quantum system $\rightarrow$ particle classes.

Worked Example: The Square

- D_4 = the 8 rigid motions that leave a square looking identical.
- Four rotations (0° , 90° , 180° , 270°) and four reflections.
- Closure, identity (0°), inverses (270° undoes 90°), associativity: all hold.
- Rotate-then-reflect $\neq$ reflect-then-rotate: groups need not be commutative.

When It Is Not a Group

- Integers under $+$: a group (identity 0, inverse of 7 is -7).
- Integers under $\times$: inverses fail ($1/7$ is not an integer). Not a group.
- Preprocessing transforms: the deciding axiom is always inverses.
- A lossy step (projection, aggregation, clipping) kills invertibility.

“Can I undo this step?” and “Do my transforms form a group?” are the same question.

Homomorphisms

- A map f between groups that preserves the operation: $f(a \cdot b) = f(a) \cdot f(b)$.
- Combine-then-map lands where map-then-combine lands.
- This is “same structure, different content” made precise.
- An *isomorphism* is a homomorphism that is also a bijection.

Isomorphism = Same Group, Different Clothes

- The 4 rotations of a square and $\{0,1,2,3\}$ under addition mod 4: isomorphic.
- Same structure, so anything true of one is true of the other.
- But same size does not mean same group.

Order-4 group	Generated by one element	Every element self-inverse
$\mathbb{Z}_4$ (cyclic)	yes	no
Klein four	no	yes

The Theme That Runs Everywhere

- The right map between structures is the one that preserves the structure.
- Group homomorphism preserves the operation.
- Linear map (Chapter 8) preserves addition and scaling.
- Continuous map (Chapter 9) preserves nearness.
- What you choose to preserve *defines* the field you are in.

Subgroups: Partial Symmetry

- A subgroup is a group sitting inside a larger group.
- The 4 rotations are a subgroup of the 8 symmetries in D_4 .
- They are the symmetries that also preserve handedness (never flip).
- Subgroups name the finer distinctions a system can make.

Quotients: Coarsen on Purpose

- Declare all elements related by a subgroup to be “the same.”
- Integers, collapse multiples of 3 $\rightarrow \mathbb{Z}/3\mathbb{Z}$, the three-position clock.
- What is discarded: which multiple of 3. What survives: the remainder.
- Subgroup zooms in; quotient zooms out. A resolution control on symmetry.

Representations

- Abstract group = what symmetries exist and how they compose.
- Representation = how they act, as matrices on a vector space.
- Composition of group elements becomes matrix multiplication.
- This is where group theory meets linear algebra (Chapter 8).

Why Spectroscopy Can Predict

- Two molecules, different chemistry, same symmetry group.
- Their vibrations form a space the group acts on: a representation.
- It splits into the group's irreducible pieces, fixed by the group alone.
- So the *pattern* of distinct frequencies is set by the group, not the chemistry.
- $SO(3)$ and continuous groups carry Noether: symmetry $\rightarrow$ conserved quantity.

The Erlangen Program

- Klein, 1872: a geometry is the study of what a transformation group leaves invariant.
- Rigid motions $\rightarrow$ distance, angle $\rightarrow$ Euclidean.
- Add scaling $\rightarrow$ ratios, parallelism $\rightarrow$ affine. Projective $\rightarrow$ straightness, cross-ratio.
- Any continuous deformation $\rightarrow$ only holes and connectivity $\rightarrow$ topology (Chapter 9).
- Bigger group $\rightarrow$ fewer invariants $\rightarrow$ coarser geometry.

The field you stand in is set by the transformations you agree to ignore.

Takeaway and Bridge

- Symmetry is now an algebra: groups, homomorphisms, subgroups, quotients, representations.
- One thesis unifies Chapters 3, 5, and 7: choosing a group is choosing a geometry.
- But representations act on a *vector space*, and we never said what that is.
- Chapter 8: what a canvas is, and why linearity lives in the canvas, not the system.